Power Analysis of Subgroup Representation in Clinical Trials: Evidence from the ACTG 175 Trial

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This study evaluates whether subgroup representation in clinical trials provides sufficient statistical power to detect meaningful treatment effects. Using the ACTG 175 trial as a case study, we examine how subgroup size and event counts influence the ability to detect heterogeneity in treatment response.

1 Introduction

Randomized clinical trials are typically powered to detect treatment effects in the overall study population rather than within demographic subgroups. This can lead to underrepresentation for female and minority ethnic groups, reducing the effective power of statistical tests and the ability to detect critical differences in treatment effects. Prior work has shown underrepresentation of women and ethnic minority groups in pivotal clinical trials across therapeutic areas (**?**; **?**; **?**).

A key feature of many clinical trials, including ACTG 320, is that outcomes are measured as time-to-event outcomes and analyzed using survival analysis methods. In this framework, statistical inference is driven by the number of observed events (e.g., disease progression or death) instead of the number of participants enrolled. As a result, subgroups with smaller sample sizes tend to yield fewer events, which can substantially reduce the power to detect treatment effects.

This distinction between sample size and event count is central to the present study. Even when treatment effects differ across demographic groups, studies may lack sufficient statistical power to detect these differences if the number of observed events is small. This creates a structural limitation in subgroup analysis that is not always apparent when focusing solely on enrollment counts.

The purpose of this paper is to use the ACTG 320 clinical trial (**?**) as a case study to investigate how underrepresentation in gender and ethnic subgroups impacts the statistical power to detect clinically meaningful treatment effects.

The rest of this paper discusses the ACTG320 data set elements and limitations, provides a brief overview of key survival analysis concepts including hazard functions, Cox proportional hazards models and how hypothesis testing and power analysis are applied in this context. The paper concludes with a discussion of the results, implications and areas for further research.

2 Data

The ACTG 320 dataset is derived from a randomized, double-blind clinical trial evaluating the efficacy of combination antiretroviral therapy in HIV-infected patients with advanced disease (CD4 cell counts 200 cells/mm^3). Participants were stratified by their CD4 cell counts at enrollment for those with CD4 cell count < 50 and patients with CD4 cell count 51 - 200 cells/mm^3). Patients in each group were randomly assigned to receive either a two-drug nucleoside regimen with a placebo or a three-drug regimen that included the protease inhibitor indinavir.

The study's primary endpoint is a time-to-event defined as either progression to AIDS or death. A total of 1,156 patients participated in the trial but only 96 experienced progression to AIDS or death during follow-up. Researchers also investigated two secondary endpoints: changes in CD40 cell levels tracked at intervals during the study and changes in plasma HIV-1 RNA concentrations.

The dataset contains participant level records that include treatment assignment, baseline clinical characteristics (e.g., CD4 count, prior antiretroviral therapy), demographic variables, and follow-up information on clinical. There are also subgroup indicators (e.g., sex and race/ethnicity), which allow for investigation of treatment effect heterogeneity and, importantly for this study, assessment of how reduced subgroup sample sizes and event counts impact statistical power.

The publicly available data set does not include data on the secondary outcomes so this paper will focus on power analysis on primary endpoint only. Considering that there 96 of enrolled participants experienced the AIDS or death, the power of any subgroup tests is expected to be low, limiting the precision of any subgroup level estimation treatment effects.

3 Methods

Clinical trials evaluating time-to-event outcomes require statistical methods that properly account for censoring and the timing of events. In the context of HIV clinical trials such as ACTG 320, the primary endpoint is the time to progression to AIDS or death, making survival analysis the appropriate framework. The section will start with a review of key concepts in survival analysis and progress to regression model formulations, hypothesis testing and power analysis definitions that are applicable.

3.1 Survival and Hazard Functions

Let T denote a non-negative random variable representing the time to the event of interest. The survival function is defined as

$$S(t) = P(T > t).$$

This quantity represents the probability that an individual remains event-free beyond time t and provides the basis for survival and hazard analysis. The hazard function is defined as

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t \mid T > t)}{\Delta t} \qquad (1).$$

where Δt is a small time interval, and $h(t)$ represents the instantaneous event rate at time t conditional on survival up to time t (**?**). Unlike the survival function, the hazard function is not a probability but determines how quickly the survival probability declines over time. Higher values of $h(t)$ correspond to a greater risk of the event occurring immediately after time t, while lower values indicate slower event occurrence. The hazard and survival functions are related by

The hazard function can be derived from (1) as the ratio of the pdf of T at time t and the survival function

$$h(t) = \frac{f(t)}{S(t)}.$$

3.2 Cox Proportional Hazards Model and Hazard Ratios

The Cox proportional hazards model are used in time-to-event clinical studies to estimate the association between covariates and the hazard of the event. The model is semi-parametric since it doesn't require specifying the baseline hazard function. For subject i with covariate vector X_i, the Cox model is

$$h_i(t \mid \mathbf{x}_i) = h_0(t) \exp(\mathbf{x}_i^\top \beta),$$

where $h_0(t)$ is the unspecified baseline hazard function and $Beta$ is a vector of regression coefficients. In the Cox model, comparisons between groups are made through ratios of hazard